TOOL DESIGN

Subject Code : 23A03504a (Professional Elective - IV)

UNIT II : DESIGN OF CUTTING TOOLS

Syllabus (as given)

Mechanics of Metal cutting - Oblique and orthogonal cutting - Chip formation and shear angle - Single-point cutting tools - Milling cutters - Hole making cutting tools - Broaching Tools - Design of Form relieved and profile relieved cutters - Design of gear and thread milling cutters.

1. Mechanics of Metal Cutting

- Metal cutting = removal of unwanted material from a workpiece in the form of chips using a wedge-shaped, harder tool.
- The tool moves relative to the work with cutting speed V, feed f and depth of cut d.
- Cutting action: the tool compresses the metal ahead of it -> metal is sheared along a narrow zone (shear plane) -> the chip slides up the rake face of the tool.
- Three deformation zones: primary shear zone (work -> chip), secondary zone (chip-tool friction) and tertiary zone (flank-work rubbing).
- Heat generated: about 75 % goes with the chip, 20 % into the tool and work, rest in the surroundings.

1.1 Orthogonal and Oblique cutting

Feature	Orthogonal (2-D) cutting	Oblique (3-D) cutting
Cutting edge	perpendicular (90 deg) to cutting velocity	inclined at angle i (inclination) to the normal
Chip flow	chip flows over rake face, perpendicular to edge	chip flows sideways at an angle to the normal
Forces	2 components (Fc and Ft)	3 components (Fc, Ft, and radial Fr)
Edge length	edge length = width of cut	edge longer than width of cut -> more cutting edge in action

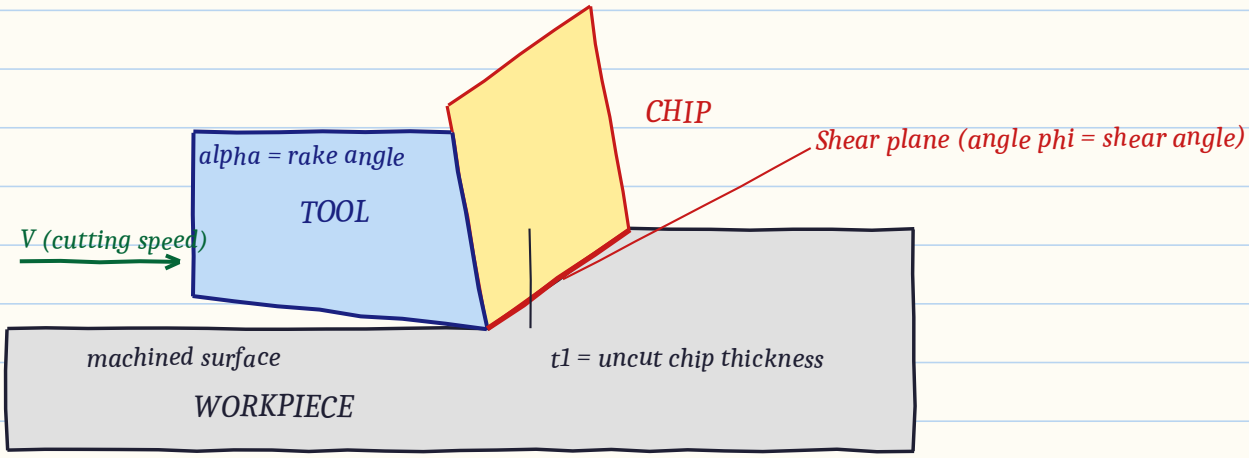
Feature	Orthogonal (2-D) cutting	Oblique (3-D) cutting
Use	theory, analysis, e.g. turning a tube end, broaching, planing	practical cutting: turning, milling, drilling
Surface	simple, easy to analyse	better finish and tool life (gradual engagement)

- Assumptions of orthogonal cutting theory (Merchant): sharp tool; 2-D deformation; continuous chip without built-up edge; uniform cutting velocity; shear occurs on one plane; width of chip = width of cut.

1.2 Chip formation and shear angle

$r = t1/t2 = \sin(\phi) / \cos(\phi - \alpha)$

$\tan(\phi) = r\cos(\alpha) / (1 - r\sin(\alpha))$



- Uncut chip thickness t1 (= feed in turning); chip thickness t2 after cutting; t2 > t1 always (chip is compressed and thickened).
- Chip thickness ratio (cutting ratio) $r = t1/t2 = \sin(\phi) / \cos(\phi - \alpha)$ ($r < 1$).
- Hence the shear angle $\tan(\phi) = r\cos(\alpha) / (1 - r\sin(\alpha))$.
- Chip compression ratio = $1/r = t2/t1$ (> 1). Larger shear angle -> thinner chip, lower forces, less heat -> better cutting .
- Shear strain in the chip: $\gamma = \cot(\phi) + \tan(\phi - \alpha) = \cos(\alpha) / (\sin(\phi)\cos(\phi - \alpha))$.
- Velocities: chip velocity $Vc = V\sin(\phi)/\cos(\phi - \alpha) = r\,V$; shear velocity $Vs = V\cos(\alpha)/\cos(\phi - \alpha)$.

Merchant's shear angle relations

- Merchant (min. energy): phi = 45 deg + alpha/2 - beta/2
- Lee and Shaffer (slip-line): phi = 45 deg + alpha - beta
- where alpha = rake angle, beta = friction angle (tan(beta) = mu = F/N)
- Positive rake or lower friction -> larger phi -> easier cutting

1.3 Types of chips

Chip type	How it forms	Conditions
Discontinuous (segmented)	metal fractures ahead of tool; small broken pieces	brittle work (cast iron, bronze), low speed, large feed, small rake
Continuous	ductile metal flows in a long ribbon; good finish	ductile metal, high speed, small feed, large rake, good lubricant
Continuous with built-up edge (BUE)	layer of work material welds to tool tip, breaks off periodically	ductile metal, low-medium speed, high friction; gives rough finish

- Chip breakers (groove or step on rake face) curl and break long continuous chips for safety.

1.4 Forces on the tool - Merchant's circle

MERCHANT FORCE CIRCLE

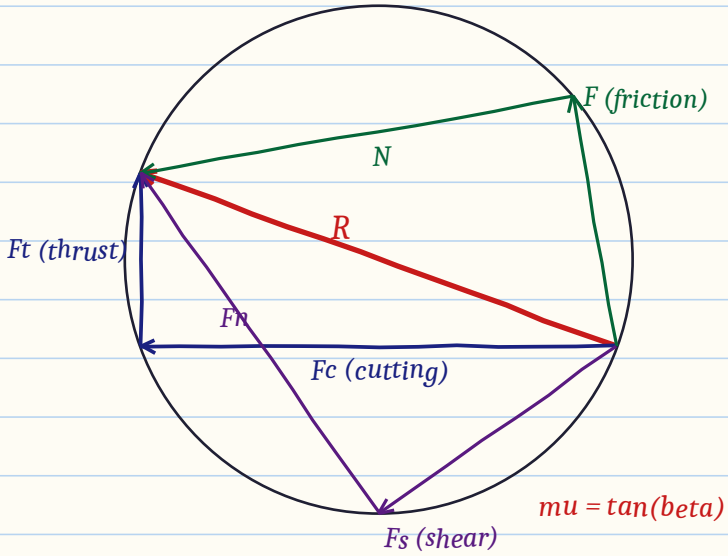
Diameter of circle = R

beta = friction angle = tan-1(F/N)

F = R sin(beta)
N = R cos(beta)

Fs = Fc cos(phi) - Ft sin(phi)
Fn = Fc sin(phi) + Ft cos(phi)

F = Fc sin(alpha) + Ft cos(alpha)
N = Fc cos(alpha) - Ft sin(alpha)



- Merchant's circle is a graphical tool to find all force components from just two measured forces (Fc and Ft from a dynamometer).
- Resultant R = sqrt(Fc2 + Ft2). Friction angle beta = alpha + tan-1(Ft/Fc).

- Friction: F = Fc sin(alpha) + Ft cos(alpha); N = Fc cos(alpha) - Ft sin(alpha); mu = F/N.
- Shear: Fs = Fc cos(phi) - Ft sin(phi); Fn = Fc sin(phi) + Ft cos(phi).
- Shear force = shear stress x shear plane area: Fs = tau_s x (b t1 / sin(phi)).
- Power: P = Fc x V (watt). Specific cutting energy = Fc/(b t1) (N/mm2); about 1.5 to 3 times for steel than for aluminium.

Solved idea (Exam)

- Given: t1 = 0.25 mm, t2 = 0.75 mm, alpha = 10 deg
- r = 0.25/0.75 = 0.333; tan(phi) = 0.333 cos10 / (1 - 0.333 sin10) = 0.328/0.942 = 0.348
- phi = 19.2 deg. Chip is much thicker -> heavy deformation

2. Single Point Cutting Tools

- Single point tool: one main cutting edge, used on lathe, shaper, planer, boring bar, slotter.
- Parts: shank (held in tool post), face / rake face (chip slides), flank (end and side), nose , cutting edges (side and end), heel .

2.1 Tool signature (nomenclature)

- ASA (American) system: back rake - side rake - end relief - side relief - end cutting edge angle - side cutting edge angle - nose radius .
- Example signature: 8 - 14 - 6 - 6 - 6 - 15 - 1 mm (HSS tool for steel).

Angle	Function / effect
Back (top) rake	controls direction of chip flow along the tool axis; +ve reduces cutting force
Side rake	main influence on shearing; +ve reduces force and heat; too large weakens edge
End relief (front clearance)	prevents rubbing of flank with machined surface (5-10 deg)
Side relief	prevents flank rubbing with work feed direction (5-10 deg)
End cutting edge angle (ECEA)	avoids end edge rubbing; 5-30 deg
Side cutting edge angle (SCEA, lead)	distributes load on a longer edge; 15-30 deg; thins chip
Nose radius	strengthens tip; better finish (0.4-1.6 mm); too big -> chatter

- Rake angle guide: hard brittle materials -> small or negative rake (carbide cast iron); soft ductile (aluminium) -> large positive rake 15-30 deg; steel (HSS) 8-14 deg.

2.2 Design of a single-point tool (shank)

- Tool shank is treated as a cantilever beam loaded by the main cutting force Fc at the tip.
- Bending moment M = Fc x L (L = overhang). For a rectangular shank b x h (h in direction of force):
  - Section modulus Z = b h2 / 6; bending stress sigma = M/Z = 6 Fc L/(b h2) <= allowable.
  - Hence h = sqrt(6 Fc L / (b x sigma_allow)). Usually h/b = 1.25 to 1.6 (b = 0.75 h for good rigidity).
- Deflection check: delta = Fc L3 / (3 E I), I = b h3/12; keep delta small (< 0.025 mm for finishing).
- Fc is estimated from: Fc = Ks x (d x f); Ks = specific cutting pressure (steel 1800-2500 N/mm2, C.I. 1000-1300, Al 600-800).
- Overhang L should be not more than 1.5 times the shank height for rigidity.
- Standard shank sizes: 12x12, 16x16, 20x20, 25x25 (square); 12x16, 16x25 (rectangular) mm; HSS bit brazed or clamped.

Tool material and construction

- Solid HSS tool bit - small tools
- Brazed tip tool - carbide tip brazed on MS shank
- Indexable insert tool - clamped throw-away carbide insert (most economical; no regrind)
- Tip is positioned with top clamp, pin lock or screw-on methods

3. Milling Cutters

- Milling = removal of metal by a multi-tooth rotating cutter while the work feeds; each tooth takes an intermittent cut.
- Classification:

1. Plain (slab) milling cutter: cylindrical, teeth on periphery only; straight or helical teeth; flat surfaces.
2. Side and face cutter: teeth on periphery and one/both sides; slots, steps.
3. Slitting saw: thin (0.5-5 mm) for cut-off, narrow slots.

4. *End mills:* teeth on end and periphery; slots, pockets, profiles; solid or inserted.
5. *Face mill:* large diameter, inserts on face; heavy flat surfacing.
6. *Angle cutters (single/double):* dovetails, V-grooves, chamfers. *T-slot cutter* and *Woodruff keyseat cutter* .
7. *Form cutters:* convex, concave, corner-rounding, gear, thread cutters.

- Another classification: *profile-sharpened* (sharpened on land behind edge) and *form-relieved* (sharpened on rake face only, profile retained).

3.1 Nomenclature and design parameters

- *Parts:* body, tooth, land, face (rake), gash, fillet radius, root, back-off, keyway, bore.
- *Angles:* radial rake (0-15 deg), clearance (3-10 deg primary, secondary 10-15 deg), helix angle (20-45 deg), land width (0.8-2 mm).

Design item	Typical value / rule	Remark
Number of teeth z	z = (0.04 to 0.06) D to ~ 2 x sqrt(D)	fewer teeth for roughing (more chip space)
Diameter D	from machine arbor, depth of cut	D = arbor dia x 2.5 to 3 (minimum)
Helix angle	20-45 deg (slab) ; 10-20 (side)	smooth cut, axial thrust to be balanced
Rake angle	HSS: 10-15 deg steel, 0-5 C.I., 25 Al	carbide: 0 to -5 deg
Clearance	HSS: 4-10 deg	carbide 6-10 deg
Face width	equal to width of work + overrun	

- *Cutting force on milling:* power P = Q x Ks ; Q = metal removal rate = w x d x feed rate (mm3/min); feed per tooth ft = f / (z N).
- *Cutting speed V = pi D N/1000 (m/min)* ; typical HSS on steel 20-30, on aluminium 150-300 m/min; carbide 3-5 times more.
- *Up (conventional) milling:* cutter rotates against feed -> chip thin to thick; needs rigid clamp. *Down (climb) milling:* with feed -> thick to thin; good finish, needs backlash eliminator.

4. Hole Making Cutting Tools

4.1 Twist drill

- Parts: point, body (flutes, land, margin, web), shank (straight or Morse taper), tang.
- Geometry: point angle 118 deg (steel, general), 90-100 for C.I., 130-140 for hard materials / plastics 60-90; helix (rake) angle 24-32 deg; lip clearance 8-15 deg; chisel edge angle 120-135 deg; web thickness = 0.12-0.2 D (thickens toward shank).
- Forces: Torque T = Ks-based: T ~ K D2 f0.8 ; thrust ~ K D f0.8; about 50-60 % of thrust is due to the chisel edge -> web thinning .
- Feed per rev: f = 0.01-0.02 D (steel); speed 20-30 m/min (HSS).
- Design points: flute length > hole depth + clearance; ratio L/D up to 5 normal; deep-hole -> gun drill with coolant through hole.
- Defects: oversize (unequal lips), walk (poor chisel), burr; fixed by correct grinding and using bushes.

4.2 Reamers

- Reamer: multi-flute tool to finish a drilled hole to accurate size and fine finish (H7, Ra 0.8-1.6). Removal is small: 0.15-0.5 mm.
- Types: hand reamer (straight/spiral flutes, square shank), machine (chucking) reamer, taper reamer, expansion/adjustable reamer, shell reamer, floating holder reamer.
- Teeth number z = 1.5 sqrt(D) + (2 to 4), even number, irregular (unequal) spacing to avoid chatter; spiral left-hand flutes for through holes with keyways.
- Design: lead (chamfer) 45 deg at the start, bevel 1-3 deg taper at the tip, margin land 0.1-0.2 mm, rake 0-5 deg, relief 5-8 deg.

4.3 Boring tools, counterbores, countersinks

- Boring bar - single point tool in a bar to enlarge a hole; short and stiff (L/D < 4) to avoid chatter.
- Counterbore - enlarges the end of the hole for bolt heads; countersink - cone 60-120 deg for flat head screws; spot facer - flat seat.
- Taps cut internal threads: hand, machine, spiral-pointed; tap drill size = major dia - pitch.

5. Broaching Tools

- Broaching uses a long multi-tooth tool pulled or pushed over/through the work; the rise per tooth does the feed; finished in a single stroke .
- Internal broach (holes, splines, keyways) and surface broach (flat, contour); very high production rate and accuracy (IT7, Ra 0.8).
- Parts of a pull broach: pull end (shank), front pilot, roughing teeth, semi-finishing teeth, finishing teeth, rear pilot, follower.
- Teeth design:
  - Rise per tooth (RPT / step) 0.01-0.1 mm; roughing 0.05-0.1, finishing 0.01-0.02, finishing teeth have no rise (sizing).
  - Pitch p = (1.76 to 2.2) x sqrt(L) (L = length of cut in mm) - for chip space; at least 2-3 teeth in cut simultaneously: p < L/2.
  - Gullet (chip space): depth h ~ 0.4 p, root radius r = 0.25 p-ish; must hold the chip curl: chip volume x 3-5 (compression).
  - Rake 5-20 deg (steel 10-15); relief 1-3 deg; land 0.1-0.5 mm.
  - Total broach length = pull end + front pilot + (roughing + semi-finishing + finishing) + rear pilot.
- Broach force: Fb = Ks x (width of cut per tooth) x (RPT) x (teeth in cut). Check tensile strength at the smallest section (the first tooth root): stress = F/A <= allowable (HSS ~ 300-400 N/mm2 safe working).
- Chip breakers (nicks staggered on teeth) for wide cuts. Material: HSS (M2, M3), or carbide tipped for high production.

Advantages / limits of broaching

- Fast, accurate, good finish, no skilled operator
- Tool is costly, only for high volume; one broach for one shape; for through shapes only

6. Design of Form Relieved and Profile Relieved Cutters

6.1 Profile (ground) relieved cutters

- Teeth are relieved by grinding the land behind the cutting edge (clearance angle) e.g. plain, side & face, end mills.
- After resharpening, the profile of the cutter changes when the tooth is sharpened on the relief land . Suitable only for simple shapes (plain, helical edges), and for standard

cutters .

- Rake angle is generally taken zero or small ; sharpened by grinding the back of the tooth, so the diameter reduces slowly.

6.2 Form relieved cutters

- Used for complex profiles (gear cutter, thread milling cutters, form tools). The tooth back is relieved by a cam (Archimedean spiral) so that the profile is kept unchanged after resharpening on the face (rake face) only .
- The relief is produced by cutting the tooth back on the lathe with a relieving attachment: radius falls uniformly with angle - an Archimedean spiral with constant drop.

Cam relief (design equations)

- Pitch angle between teeth = 360/z; drop (cam rise) K = (pi D / z) x tan(alpha_c) (alpha_c = clearance angle, 10-12 deg for HSS)
- Cam lift K = (pi D / z) tan(alpha); standard rise adopted as the next higher cam rise (e.g. 1.5, 2, 3, 4, 5 mm ...)
- Reduction in diameter after full life = (back-off x number of regrinds)
- Radial depth of tooth H = profile depth + (0.5 to 1.5 mm) clearance + 1/2 K + fillet radius r; H and tooth pitch fix fillet radius (r = 0.25 H)
- Tooth thickness at root ~ 0.5 to 0.6 x pitch; body thickness D - 2H for the bore + keyway

- Relief angle on the sides of the profile: sides also need clearance; for profile slopes at angle theta to the axis, the side clearance angle alpha_s = tan-1(tan(alpha) x sin(theta)); minimum 1-2 deg is kept.
- Why Archimedean spiral? So that the radius reduces linearly with angle and the clearance angle stays the same after repeated grinding.

6.3 Design steps for a form relieved cutter

1. Fix the profile to be produced and the machine arbor.
2. Select outside diameter D , bore d (arbor), no. of teeth z.
3. Select rake angle (0 for form cutters, so profile preserved), clearance (alpha = 8-12 deg).
4. Calculate cam rise K from alpha, z, D; check and adjust (Standard cam).
5. Decide tooth height H (profile depth + clearance + K + fillet) and fillet radius r.
6. Check that root diameter D - 2H > bore dia + (6 to 10 mm) for keyway and strength; adjust z and D.

7. Prepare drawing with all dimensions, tolerance, hardness 62-64 HRC (HSS).

7. Design of Gear Milling Cutters

- Gear teeth by form milling : a form-relieved (involute) cutter takes the shape of the tooth space; indexing head turns the gear blank after each space.
- The involute shape changes with number of teeth , so cutters are supplied in a set of 8 cutters (No. 1 to 8) for a given module and pressure angle; each cutter is exact for the smallest tooth number in its range, slightly thin tooth space for others.

Cutter no.	Teeth range	Cutter no.	Teeth range
1	135 to rack	5	21 to 25
2	55 to 134	6	17 to 20
3	35 to 54	7	14 to 16
4	26 to 34	8	12 to 13

- Design data (Module m, pressure angle 20 deg): addendum = m; dedendum = 1.25 m; tooth depth h = 2.25 m; circular pitch p = pi m; space width at pitch circle = pi m/2; pitch circle dia = m z.
- Cutter dimension: cutter tooth thickness at pitch = (pi m/2) - small allowance; depth of cut = 2.25 m (cutter height > 2.25 m).
- Cutter outside dia chosen from standard (e.g. m = 1: 50 mm; m = 5: 100 mm...). Cutter has involute form for the smallest tooth number in range , using the base circle Rb = (m z/2) cos(phi).
- Involute profile coordinates : x = Rb (sin theta - theta cos theta), y = Rb(cos theta + theta sin theta) - generated point by point for the profile table.
- Alternatives: hob (continuous generating, better accuracy), gear shaper cutter , rack cutter .

8. Design of Thread Milling Cutters

- Thread milling cutters cut external / internal threads on a thread milling machine (single-pitch or multiple-pitch cutter).
- Types: (a) single-form cutter (one-tooth, like a disc with thread form; for large pitch, e.g. worms, acme, trapezoid) and (b) multiple-form (many-ring) cutter (several rings, as

wide as thread length; cuts the full thread in about 1.2 rev of the work).

- Principle: cutter rotates at high speed on a slightly inclined axis (helix angle of thread, lambda = tan-1(lead / pi d)); work rotates slowly and the cutter moves axially one lead per work rev.

Design data

- Lead L = pitch x starts; helix angle lambda: tan(lambda) = L/(pi d_m) (d_m = mean dia)
- Cutter inclination = lambda; cutter profile differs from the thread normal section (corrected for helix angle)
- Cutter outside dia ~ 60-100 mm (single), 20-60 for multi-ring; tooth form angle 60 deg (metric, ISO) or 29 deg (acme)
- Cutter is form relieved (back-off by cam) so that profile stays after face sharpening
- Number of teeth 10-16 per rev; rake 0 deg for form accuracy

- Profile correction : because the cutter axis is inclined, thread space measured normal to the helix is slightly different from the axial section; for large helix angle use a corrected profile .
- Thread milling gives high accuracy at high production for lead screws, worms; internal thread milling can be done by CNC thread mill (single point, circular interpolation) .

9. Tool Wear, Tool Life and Cutting Fluids

- Tool wear types: flank wear (on the clearance face; VB = 0.3 mm limit for HSS/carbide finishing), crater wear (on rake face by chip; depth KT), notch wear (at depth-of-cut line), nose wear , edge chipping and thermal cracking.
- Causes: abrasion, adhesion (welding and tear), diffusion (carbide at high temperature), oxidation, fatigue from interrupted cuts.
- Tool life criteria: flank wear VB = 0.3 mm (uniform) or 0.6 mm (non-uniform); crater depth; surface finish deterioration; force rise; failure of the edge.

Taylor's tool life equation

- V T^n = C (V = cutting speed m/min, T = tool life in min)
- n = 0.1-0.15 HSS; 0.2-0.4 carbide; 0.4-0.6 ceramic
- Extended: V T^n f^a d^b = C (feed and depth also reduce life)
- Speed has the largest effect: raise V by 20 % -> life is cut roughly by half (HSS)

- Example: HSS tool, V = 30 m/min gives T = 60 min. n = 0.125. Find T at V = 36 m/min.

(30/36)^(1/0.125) = (0.833)^8 = 0.233 -> T = 60 x 0.233 = 14 min .

- Economic tool life (min cost): T_opt = (1/n - 1) x (Tc + Cc/Co) ; Tc = tool change time, Cc = tool cost per edge, Co = machine + labour rate per minute.
- Factors improving tool life: correct geometry, proper material grade, rigid set-up, coolant, reduce speed, avoid interrupted cut, sharp edge and honing.

9.1 Cutting fluids

- Functions: cooling (carry away heat), lubrication (reduce friction), washing away chips, corrosion protection, better finish.
- Types: neat oils (low speed, heavy cut - broaching, tapping), soluble (emulsion) oils (general high speed; mostly cooling), synthetic, semi-synthetic, gases / air (dry machining, MQL).
- Not used in milling of cast iron (dry or air), and with ceramic tools (thermal shock).

10. Worked Examples

Example 1 - Single point tool shank

- A tool has Fc = 2400 N, overhang L = 40 mm, shank width b = 16 mm, allowable bending stress 160 N/mm2. Find the shank height h.
- h = sqrt(6 Fc L/(b sigma)) = sqrt(6 x 2400 x 40 /(16 x 160)) = sqrt(225) = 15 mm .
- Use standard 16 x 16 mm square shank (h = 16 > 15, so safe).

Example 2 - Turning power

- Turning steel with d = 3 mm, f = 0.25 mm/rev, Ks = 2000 N/mm2, V = 60 m/min.
- Fc = Ks x d x f = 2000 x 3 x 0.25 = 1500 N .
- Power P = Fc V/60 = 1500 x 60 / 60 = 1500 W = 1.5 kW at the tool; for machine efficiency 80 % -> motor power = 1.88 kW.

Example 3 - Plain milling cutter data

- Cut a 40 mm wide slab, 4 mm deep, in steel using D = 80 mm, z = 8, N = 100 rpm, table feed 120 mm/min.
- Cutting speed V = pi x 80 x 100/1000 = 25.1 m/min .
- Feed per tooth ft = 120 /(8 x 100) = 0.15 mm/tooth .
- Metal removal rate Q = w x d x feed = 40 x 4 x 120 = 19200 mm3/min = 320 mm3/s.
- Power at cutter = Q x Ks = 320 x 2000 N.mm/s = 640 000 N.mm/s = 640 W (Ks = 2000 N/mm2 for steel).

Example 4 - Broach pitch

- Length of cut L = 40 mm; rise per tooth 0.04 mm. Pitch p = (1.76 to 2.2) x sqrt(40) = 11.1 to 13.9 mm -> take p = 12 mm .
- Check at least 2 teeth in cut: L/p = 40/12 = 3.3 >= 2 -> OK.
- Number of roughing teeth = (total stock to be removed)/RPT = 1.2/0.04 = 30 teeth.

Example 5 - Cam rise of a form relieved cutter

- Cutter D = 75 mm, z = 10, clearance angle 10 deg.
- K = (pi D/z) tan(alpha) = (pi x 75/10) x tan 10 = 23.56 x 0.1763 = 4.15 mm -> choose standard cam rise 4 to 5 mm (take 4.5 mm, adjust relief angle accordingly).

Common viva questions

- Why is shear angle important? What affects it?
- Difference between form-relieved and profile-relieved cutters?
- Why does drill need web thinning? Why is reamer pitch irregular?
- Why are gear cutters supplied in sets of 8?
- Why must broach pitch be selected carefully? (chip space, 2-3 teeth in cut)

11. Quick Revision - Unit II

- Orthogonal: 2 forces; oblique: 3 forces. r = t1/t2 = sin(phi)/cos(phi-alpha).
- Merchant: phi = 45 + alpha/2 - beta/2. Merchant circle gives F, N, Fs, Fn from Fc, Ft.
- Single point tool: shank is a cantilever: h = sqrt(6 Fc L/(b sigma)).
- Milling cutters: plain, side-face, end mill, form; hole-making: drill, reamer, tap; broach: RPT, pitch = (1.76-2.2) sqrt(L).
- Form relieved cutters keep profile by cam relief (Archimedean spiral); gear cutters come in a set of 8; thread cutters inclined by helix angle.